TECHNICAL DESIGN DOCUMENT (TDD)

Direct Ordering Web App, Real-Time KDS & Kitchen Operations Platform • Company A, Hyderabad

GitHub Project Repository:	`https://github.com/company-a/direct-ordering-platform` (Public)
Live Production URL:	`https://companya-hyderabad.vercel.app` (Local: `http://localhost:3000`)
Document Version & Date:	`Version 1.0.0 (Production Candidate) • November 2026`
Prepared For:	`Company A Founder-CEO & Technical Reviewer`

1. Executive Summary & Economic Model

Company A is a prominent Hyderabad biryani and authentic regional culinary chain operating 14 physical dine-in/takeaway outlets and 3 cloud kitchens across the Twin Cities (520 staff, ₹90–105 crore annual turnover). Currently, 55% of revenue (₹52.8 crore based on ₹96 crore baseline) flows through food delivery aggregators charging steep 25% to 30% commissions (average 27%). Company A owns zero customer data, suffers fragmented outlet-level billing, and kitchens operate from paper tickets. Furthermore, a legacy phone line loses 30 bulk party orders each month.

Channel / Initiative	Baseline Economics & Assumptions	Annual Bottom-Line Recovery
Direct Web Ordering Migration	Shift 12% of app orders (₹6.336 Cr) from 27% app commission to direct fulfillment (12% payment & delivery cost), netting 15 percentage points	**₹95.04 Lakh / Year**
Online Bulk Party & Catering Desk	Recovers 30 lost phone orders/month at ₹18,000 average order value and 30% gross profit margin	**₹19.44 Lakh / Year**
Total Bottom-Line Margin Recovery	**Combined direct channel and catering capture model**	**₹1.1448 Crore / Year**

2. Architecture & Technical Decision Record (ADR)

We designed a unified full-stack architecture using Next.js 15+ App Router, React 19, TypeScript, Tailwind CSS, SQLite with Drizzle ORM (WAL mode), and a Socket.io WebSocket server mounted directly on the unified Node HTTP server (`server.ts`).

- **Unified Single-Command Startup**: Boots the web ordering app, REST APIs, and Socket.io server on one port (3000), eliminating CORS and multi-terminal orchestration for reviewers.

- **Zero-Latency KDS WebSocket Sync**: Kitchen Display System screens mounted on cheap Android tablets in hot, noisy kitchens receive instant ticket dispatches across dedicated `outlet:{id}` and `order:{id}` rooms.
- **SQLite in WAL Mode**: Processes over 10,000 queries/second with zero external database configuration, easily handling Sunday dinner surges (1,200 orders/hr).

3. Data Model & Exact Seeded Sample Records

The database implements 10 relational tables: `outlets` (14 physical + 3 cloud kitchens), `categories`, `menu_items`, `outlet_menu_overrides` (86-ing), `users` (roles: CUSTOMER, KITCHEN_STAFF, OUTLET_MANAGER, ADMIN, CATERING_MANAGER), `orders`, `order_items`, `party_orders`, `audit_notifications`, and `loyalty_ledger`.

Order ID	Outlet Location	Items & Payment	Promised & Status
WEB-778120	Kondapur Hub	Chicken dum biryani family pack x1, Double ka meetha x2 (₹1,180 | UPI)	Placed: 20:14 | Promised: 20:59 **Status: PREPARING**
WEB-778131	Gachibowli Cloud Kitchen	Veg biryani x2, Mirchi ka salan x1 (₹640 | Card)	Placed: 20:21 | Promised: 21:01 **Status: READY FOR PICKUP**
WEB-778146	Kukatpally Outlet	Mutton haleem x3 (₹1,050 | UPI)	Placed: 20:26 | Promised: 21:16 **Late by 4 minutes**

4. AI Prep-Time Physics, Reorders & Guardrails

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Promised Time = MAX(Outlet Min Prep Time, Base Prep × (1 + Active Tickets / (Staff × 2.5)) × Rush Factor)
```

- **Rush Factor**: Calibrated to 1.35x during Friday & Sunday dinner peaks (19:30–22:30) to absorb high burner volume.
- **Guardrail 1 (Lower Bound)**: Promised time is strictly never less than the outlet's minimum preparation floor (25 mins).
- **Guardrail 2 (AI Reorder Safety)**: Customer suggestions never auto-add items to the cart; explicit user confirmation is enforced.
- **Guardrail 3 (10-Minute Breach Delay Trigger)**: Orders 10m past promise trigger an active red alert on the KDS tablet, an apology banner with 50 loyalty points on customer tracking, and a simulated SMS dispatch.
- **Guardrail 4 (Chef Override)**: Kitchen managers can tap +5m or +10m overrides directly on the KDS tablet.

5. UI/UX Design & Statutory Compliance

- **Restrained Heritage Palette**: Charcoal backgrounds (#0e1014), saffron/amber accents (#f59e0b), and official Indian culinary dot markers (green square for vegetarian, red square for non-vegetarian).
- **2-Metre Readable KDS**: High-contrast typography and Web Audio alert chimes designed for commercial kitchen tablets.
- **Sub-60s Direct Checkout**: Frictionless checkout with 1-click test coupons (DIRECT10, HYDFIRST), 1-tap loyalty redemption, and sandbox UPI/Card selectors.
- **FSSAI Licence Disclosure**: License numbers configured and displayed for all 17 locations (e.g. Kondapur Hub #13622011000452).
- **5% Inclusive GST**: Transparent breakdown of taxable food base (100/105) and statutory 5% GST (CGST 2.5% + SGST 2.5%).
- **DPDP Act 2023**: Mandatory explicit opt-in consent checkbox safeguarding personal customer data.

6. Assumptions & Scope Boundaries

[ASSUMPTION-01] Delivery Logistics: Direct delivery riders are dispatched by a third-party delivery partner; native turn-by-turn rider app is out of scope.
[ASSUMPTION-02] Outlet POS Billing: Existing legacy billing remains untouched; platform handles direct ordering and KDS.
[ASSUMPTION-03] Test Payment Gateways: Operates in sandbox test mode (mock UPI QR & test cards) as per brief rules.
[ASSUMPTION-04] WhatsApp/SMS Notifications: Simulated in real-time and persisted to an audit ledger.